

3.4.3 Relational, Logical and Membership Operators

ریشنل آپریٹرز (Relational Operators) دو ویلیوز کا موازنہ کر کے True یا False دیتے ہیں۔

CS XI

Chapter 3: Programming Fundamentals

Lecture 11 of 24

★ Golden

3.4.3 Relational, Logical and Membership Operators ★ Golden

LEARN IT

3. Relational (comparison) operators compare two values and return a **Boolean** (True or False). They are used in conditions.

Op	Description	Example (a = 10, b = 5)	Result
=	Equal to	a == b	False
≠	Not equal to	a != b	True
>	Greater than	a > b	True
<	Less than	a < b	False
≥	Greater than or equal to	a ≥ 10	True
≤	Less than or equal to	a ≤ b	False

```
a = 10
b = 5
print(a == b)
print(a != b)
print(a > b)
print(a < b)
print(a ≥ 10)
print(a ≤ b)
```

OUTPUT

```
False
True
True
False
True
False
```

4. Logical operators combine several conditions:

A	B	A and B	A or B	not A
False	False	False	False	True
False	True	False	True	True
True	False	False	True	False
True	True	True	True	False

- `and` — True **only if both** conditions are true.
- `or` — True if **at least one** condition is true.
- `not` — **reverses** the result (True becomes False, False becomes True).

```
a = 10
b = 5
print(a > 5 and b > 5)
print(a > 5 or b > 5)
print(not a > 5)
```

OUTPUT

```
False
True
False
```

5. Membership operators check whether a value exists inside a sequence (such as a string or list): `in` and `not in`.

```
name = "Python"
print("P" in name)
print("z" not in name)
print("p" in name)
marks = [85, 90, 75]
print(90 in marks)
```

OUTPUT

```
True
True
False
True
```

"p" in name is False because strings are case-sensitive: the word has a capital P.

WORKED EXAMPLE

Real-life conditions. A person can apply for a CNIC if they are 18 or older **and** have the documents. Marks are invalid if they are below 0 **or** above 100.

```
age = 20
has_cnic = True
print(age ≥ 18 and has_cnic)
marks = 45
print(marks < 0 or marks > 100)
```

OUTPUT

```
True
False
```

Step by step: $20 \geq 18$ is True, `has_cnic` is True, True and True = **True**. For marks: $45 < 0$ is False, $45 > 100$ is False, False or False = **False** (the marks are valid).

REMEMBER IT

AND is strict, OR is generous. AND is like a teacher who needs both homework and attendance; OR is like a shop that accepts cash **or** card. Order of evaluation: `not` first, then `and`, then `or`.

COMMON MISTAKE

Do not confuse `=` (assignment, "store") with `==` (comparison, "is it equal?"). Writing `if x = 5:` is a **syntax error**; the correct form is `if x == 5:`.

اردو میں سمجھیں

ریشنل آپریٹرز (Relational Operators) دو ویلیوز کا موازنہ کر کے True یا False دیتے ہیں۔ لاجیکل آپریٹرز میں `and` اور `or` تب True دیتا ہے جب دونوں شرطیں درست ہوں، `not` تب جب کم از کم ایک درست ہو، اور `not` نتیجہ الٹ دیتا ہے۔ `in` دیکھتے ہیں کہ کوئی چیز اسٹرنگ یا لسٹ میں موجود ہے یا نہیں۔

EXAM TIP

Typical: "Evaluate `a > b` and `b > 3` for `a = 10, b = 5`" or "Write the truth table of `and` / `or`". Write each sub-condition's value first, then combine. Remember: `==` compares, `=` assigns.

CHECK YOURSELF

1. Predict the output:

```
print(True and False or True)
print(not (5 > 3) or 2 == 2.0)
```

2. Find the error:

```
x = 5
if x = 5:
    print("five")
```

Answers: 1. True and True. (True and False = False; False or True = True. not True = False; 2 == 2.0 is True; False or True = True.) 2. = is assignment; a comparison needs ==. Python reports SyntaxError: invalid syntax. Maybe you meant '==' or ':=' instead of '='? Fix: if x == 5:

Next lecture: 3.4.3 Bitwise Operators